

Lesson 3: Data Structures

- A data structure is a way of organizing data so that it can be used efficiently. Unlike an array, a linked list does not store its elements in contiguous memory.

3.1 Stacks

- A stack is a linear data structure that follows the Last In, First Out principle. For example, a pile of plates is a stack. Stacks are used to implement the undo feature in text editors.
- Push adds an element to the top of the stack.
- Pop removes the element at the top of the stack.

3.2 Queues

- A queue is a linear data structure that follows the First In, First Out principle. For example, a line of customers at a bank is a queue. Queues are used to schedule jobs sent to a printer.

3.3 Algorithms and Complexity

- An algorithm is a finite set of well-defined steps for solving a problem. The steps of problem solving are understanding the problem, planning a solution, coding the program and testing the result. Poor planning leads to programs that are hard to maintain.
- Binary search is faster than linear search because it halves the search space at each step. The average of a list is computed as $A = S / n$, where S is the sum of the values and n is the number of values.